Chapter 1

Propositional Logic

Propositional Logic

- Logic contains rules and techniques to formalize statements, to make them precise.

- Logic appears in different areas of computer science, such as programming, circuits, articial intelligence and databases.

Proposition

A statement that is **either true or false** but not **both** is called a proposition.

- Examples of propositions
  - "$1 + 1 = 2$" $\cdots$ True
  - "$1 + 1 > 3$" $\cdots$ False
  - "Singapore is in Europe." $\cdots$ False
- Examples (which are not propositions)
  - "$1 + 1 > x$" $\cdots$ ×
  - "What a great book!" . . . ×
  - "Is Singapore in Asia?" . . .×

Symbolic Logic

- Use symbols to represent statements (both have *the same truth values*)
- Use logical operators to combine statements
  - Compound propositions = propositions combined with logical operator(s)

- Three basic operators
  - $\wedge$: conjunction (and)
  - $\vee$: disjunction (or)
  - $\neg$: negation (not, alternatively ~)

Negation ¬

- Negation (not) of p: ¬p (~ p is also used)

p	¬p
T	F
F	T

Truth Table

- p: You shall pass

¬p: You shall not pass

Disjunction V

- Disjunction (or) of p with q: pVq

p	q	pVq	qVp
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

True when 'at least one' of them is true

- p V q ≡ q V p, i.e. operator V commutes

means "equivalent"

Conjunction $\wedge$

- Conjunction (and) of p with q: $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

True only when `both' of them are true

- $\wedge$ is also commutative: $p \wedge q \equiv q \wedge p$

Exclusive or

TABLE 4 The Truth Table for the Exclusive Or of Two Propositions.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Definition: Let p and q be propositions. The ***exclusive or*** of p and q, denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

De Morgan's Law

$$\neg (p \wedge q) \equiv \neg p \vee \neg q$$
$$\neg (p \vee q) \equiv \neg p \wedge \neg q$$

p q	$\neg p$	$\neg q$	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T T	F	F	T	F	F
T F	F	T	F	T	T
F T	T	F	F	T	T
F F	T	T	F	T	T

Contradiction

A statement that is always false is called a contradiction.

Example: This course is easy `and' this course is not easy

$$p \wedge (\neg p) \equiv F$$

Tautology

An expression that always gives a true value is called a tautology .

Example: $p \vee (\neg p) \equiv T$

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Always true!

Logical equivalence

- Compound propositions that have the same truth values in all possible cases are called **logically equivalent**

The compound propositions p and q are called *logically equivalent* if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ denotes that p and q are logically equivalent.

Equivalent Expressions

Consider the following three statements

- Alice is not married but Bob is not single

  $\neg h \wedge \neg b$

- Bob is not single and Alice is not married

  $\neg b \wedge \neg h$

- Neither Bob is single nor Alice is married

  $\neg(b \vee h)$

- These three statements are equivalent

  $\neg h \wedge \neg b \equiv \neg b \wedge \neg h \equiv \neg(b \vee h)$

Equivalent Expressions

The three statements

$$\neg h \wedge \neg b \equiv \neg b \wedge \neg h \equiv \neg(b \vee h)$$

are equivalent.

b h	$\neg b$	$\neg h$	$b \vee h$	$(\neg h \wedge \neg b)$	$(\neg b \wedge \neg h)$	$\neg(b \vee h)$
T T	F	F	T	F	F	F
T F	F	T	T	F	F	F
F T	T	F	T	F	F	F
F F	T	T	F	T	T	T

Logical Equivalences

- Useful laws to transform one logical expression to an equivalent one.
- Axioms ($T \equiv$ tautology, $C \equiv$ contradiction):

$\neg T \equiv F$ $\neg F \equiv T$ $\neg T \equiv C \equiv F$ $\neg C \equiv T \equiv T$

- De Morgan:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$
$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

- Commutativity:

$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$

Logical Equivalence Laws

- double negation: $\neg(\neg p) \equiv p$
- idempotent: $p \wedge p \equiv p$ and $p \vee p \equiv p$
- absorption: $p \vee (p \wedge q) \equiv p$ and $p \wedge (p \vee q) \equiv p$

Conditional Statement $\rightarrow$

Known operators: $\wedge$ conjunction (and), $\vee$ disjunction (or), $\neg$ negation.

- if p then q: $p \rightarrow q$

By definition, when p is false, $p \rightarrow q$ is true. This is called *vacuously true* or *true by default*.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Conversion Theorem

Theorem: $p \rightarrow q \equiv \neg p \vee q$

Proof

- $p \rightarrow q$ means $\neg(p \wedge \neg q)$ (it cannot be that p is true but q is false).
- Apply DeMorgan's law

p q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
T T	T	F	T
T F	F	F	F
F T	T	T	T
F F	T	T	T

Given the conditional statement if $p \to q$, it can be modified in three different ways to give raise to some other statements:

- The converse of $p \to q$ is $q \to p$.
- The inverse of $p \to q$ is $\neg p \to \neg q$.
- The contrapositive of $p \to q$ is $\neg q \to \neg p$.

Theorem 3. *We have*

$$\neg q \to \neg p \equiv p \to q,$$

that is the contrapositive of $p \to q$ is equivalent to $p \to q$!

Proof. Recall the Conversion Theorem: $p \to q \equiv \neg p \vee q$. Let us start with $\neg q \to \neg p$:

$$\begin{aligned}
&\neg q \to \neg p && \\
&\equiv \neg(\neg q) \vee \neg p && \text{Conversion Theorem} \\
&\equiv q \vee \neg p && \text{double negation} \\
&\equiv \neg p \vee q && \text{commutativity} \\
&\equiv p \to q && \text{Conversion Theorem.}
\end{aligned}$$

Another proof using truth tables is found in Exercise 13. $\square$

Only If

The contrapositive of $p \to q$ is also called only if, which is our 5th logic operator ($\triangleq$ means "equal by definition")

$$\text{only if} \triangleq \neg q \to \neg p,$$

even though it is equivalent to $p \to q$.

Example 14. The formulas are saying

$$(p \text{ only if } q) = (\neg q \to \neg p) \equiv (p \to q).$$

Let us see how this translates into sentences:

$$\underbrace{\text{“Bob pays taxes}}_{p} \text{ only if } \underbrace{\text{his income is more than 1000 SD”}}_{q}.$$

We notice where is the “only if”. Then $\neg q \to \neg p$ is

$$\underbrace{\text{“if Bob's income is less than 1000 SD,}}_{\neg q} \text{ then } \underbrace{\text{he does not pay taxes.}}_{\neg p}$$

Finally $p \to q$ is

$$\underbrace{\text{“if Bob pays taxes,}}_{p} \text{ then } \underbrace{\text{his income is more than 1000 SD.}}_{q}$$

Sufficient and Necessary Conditions

When $p \rightarrow q$, p is called a *sufficient* condition for q , q is a *necessary* condition for p.

- Being Japanese is a sufficient condition for being Asian.

  ≡ if someone is Japanese then s/he will be an Asian

- Being Asian is a necessary condition for being Japanese.

  ≡ 'if someone is not Asian, he can not be Japanese'

Example

- Let f: ‘you fix my ceiling’, p: ‘I pay my rent’.
  - ‘you fix my ceiling or I won’t pay my rent’
    - $f \vee \neg p \equiv p \rightarrow f$
  - `If you do not fix my ceiling then I won’t pay my rent’
    - $\neg f \rightarrow \neg p \equiv p \rightarrow f$
  - ‘I will pay my rent only if you fix my ceiling’
    - $\neg f \rightarrow \neg p \equiv p \rightarrow f$

Biconditional $\leftrightarrow$

- The biconditional of p and q: $p \leftrightarrow q \triangleq (p \rightarrow q) \wedge (q \rightarrow p)$
  - True only when p and q have identical truth values
- *If and only if (iff)*

p q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q$
T T	T	T	T
T F	F	T	F
F T	T	F	F
F F	T	T	T

Known operators:

$\wedge$ conjunction (and),

$\vee$ disjunction (or),

$\neg$ negation,

$\rightarrow$ conditional (if then)

Operator Precedence

- From high to low: ( ) , $\neg$, $\wedge\vee$, $\rightarrow$, $\leftrightarrow$
- When equal priority instances of *binary connectives* are not separated by (), the *leftmost one has precedence*. E.g. $p \rightarrow q \rightarrow r \equiv (p \rightarrow q) \rightarrow r$
- When instances of $\neg$ are not separated by (), the *rightmost one has precedence*:

  E.g. $\neg\neg\neg p \equiv \neg(\neg(\neg p))$

Example

- Show that $p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

$p \vee q \rightarrow r$

$\equiv (p \vee q) \rightarrow r$ (operator precedence)

$\equiv \neg(p \vee q) \vee r$ (why?)

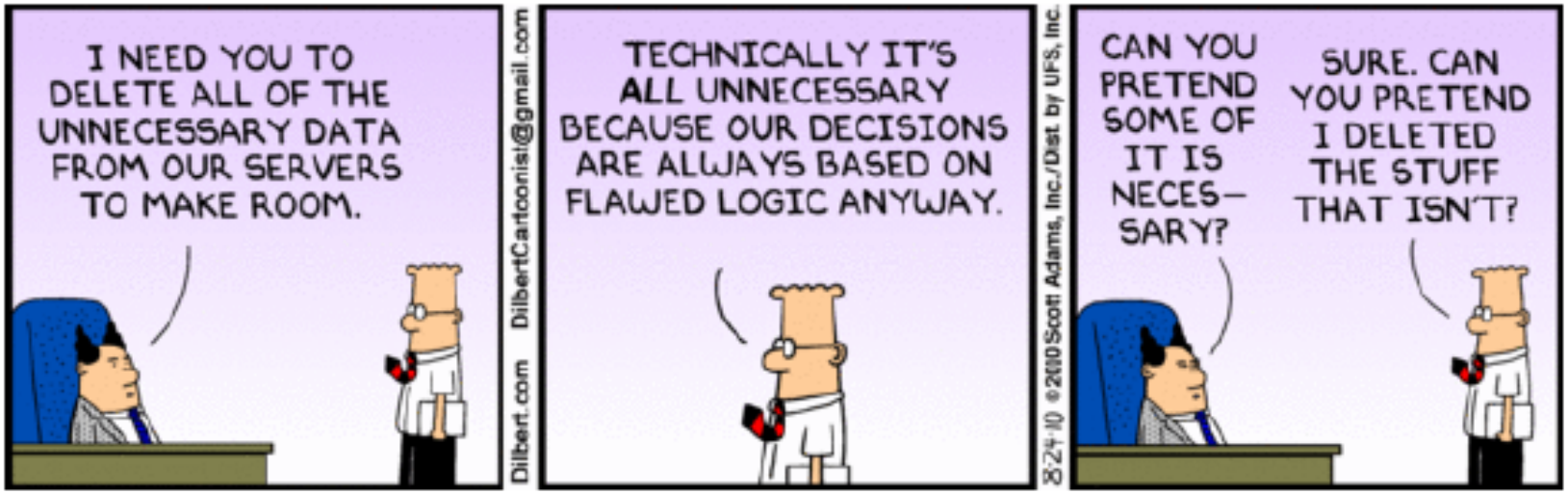
$\equiv (\neg p \wedge \neg q) \vee r$ (DeMorgan's)

$\equiv (\neg p \vee r) \wedge (\neg q \vee r)$ (why?)

$\equiv (p \rightarrow r) \wedge (q \rightarrow r)$ (why?)

Summary

- Useful logical equivalence laws
  - Proving equivalence using these laws
- Conditional & Biconditional statements
  - Sufficient and necessary conditions
- Operator precedence



Summary

1. Logic operator:

$\wedge$	conjunction (and)	$p \wedge q$
$\vee$	disjunction (or)	$p \vee q$
$\neg$	negation (not)	$\neg p$
$\rightarrow$	conditional (if then)	$p \rightarrow q$
	only if	p only if q
$\leftrightarrow$	biconditional (if and only if)	$p \leftrightarrow q$

2. Operator precedence: $(\,)$, $\neg$, $\wedge\vee$, $\rightarrow$, $\leftrightarrow$

3. Important logical equivalences

$$\neg(p \vee q) \equiv \neg p \wedge \neg q, \quad \neg(p \wedge q) \equiv \neg p \vee \neg q,$$

$$p \rightarrow q \equiv \neg p \vee q.$$

TABLE 6 **Logical Equivalences.**

Equivalence	*Name*
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

TABLE 7 **Logical Equivalences Involving Conditional Statements.**

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

Example: Show that

$$\neg(p \vee (\neg p \wedge q)) \text{ and } \neg p \wedge \neg q$$

are logically equivalent by developing a series of logical equivalences.

$\neg(p \vee (\neg p \wedge q))$	$\equiv \neg p \wedge \neg(\neg p \wedge q)$	by the second De Morgan law
	$\equiv \neg p \wedge [\neg(\neg p) \vee \neg q]$	by the first De Morgan law
	$\equiv \neg p \wedge (p \vee \neg q)$	by the double negation law
	$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the second distributive law
	$\equiv \mathbf{F} \vee (\neg p \wedge \neg q)$	because $\neg p \wedge p \equiv \mathbf{F}$
	$\equiv (\neg p \wedge \neg q) \vee \mathbf{F}$	by the commutative law for disjunction
	$\equiv \neg p \wedge \neg q$	by the identity law for $\mathbf{F}$

- Example:

Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

$(p \wedge q) \rightarrow (p \vee q)$	$\equiv \neg(p \wedge q) \vee (p \vee q)$	by Example 3
	$\equiv (\neg p \vee \neg q) \vee (p \vee q)$	by the first De Morgan law
	$\equiv (\neg p \vee p) \vee (\neg q \vee q)$	by the associative and commutative laws for disjunction
	$\equiv \mathbf{T} \vee \mathbf{T}$	by Example 1 and the commutative law for disjunction
	$\equiv \mathbf{T}$	by the domination law

Propositional Satisfiability

- **Definition:** A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true.
- When the compound proposition is false for all assignments of truth values to its variables, the compound proposition is **unsatisfiable**.
- When we find a particular assignment of truth values that makes a compound proposition true, we have shown that it is satisfiable; such an assignment is called a **solution** of this particular satisfiability problem.
- To show that a compound proposition is unsatisfiable, we need to show that *every* assignment of truth values to its variables makes it false.

- Example:

Determine whether the compound propositions

$$(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

is satisfiable

Valid Argument

An argument is a sequence of statements. The last statement is called the conclusion, all the previous statements are premises (or assumptions/ hypotheses).

A valid argument is an argument where the *conclusion is true if the premises are true.*

Valid Argument

A valid argument is an argument where the *conclusion is true if the premises are true.*

- A series of statements form a valid argument if and only if `the conjunction of premises implying the conclusion' is a tautology

- ((premise)∧(premise)) → conclusion

Valid Argument Template

$$\left.\begin{array}{l} p \to q; \\ \quad p; \end{array}\right\} \text{premises}$$

$$\therefore \; q \qquad \text{conclusion}$$

- By definition, a valid argument satisfies: "If the premises are true, then the conclusion is true"
- Also: $((p \to q) \wedge p) \to q$ is a tautology.

truth values of premises and of the conclusion

critical rows are rows with *all premises true*

if in all critical rows the conclusion is true, then the *argument is valid* (otherwise it is *invalid*).

p q	$p \to q$	$(p \to q) \wedge p$	$((p \to q) \wedge p) \to q$
T T	T	T	T
T F	F	F	T
F T	T	F	T
F F	T	F	T

No need to calculate

Counter Example

If in all critical rows the conclusion is true, then the *argument is valid* (otherwise it is *invalid).*

A *critical row with a false conclusion* is a *counter example* that invalidates the argument (=makes the argument not valid).

- A counter example indicates a situation where the conclusion does not follow from the premises.

Example 15. Consider the argument:

$$\boxed{\begin{array}{l} S = (f \wedge a \rightarrow r); \\ f; \\ \neg a; \\ \therefore \neg r; \end{array}}$$

Check whether this argument is valid.

Invalid Argument Example

'if it is falling *and* directly above me then I'll run '

'It is falling'

'it is *not* directly above me

} premises

'I will *not* run' } conclusion

$S=(f \wedge a \rightarrow r);$
$f;$
$\neg a;$ } premises

$\therefore \neg r$ conclusion

a	r	f	$\neg a$	$f \wedge a$	S	$\neg r$
T	T	T	F	T	T	F
T	T	F	F	F	T	F
T	F	T	F	T	F	T
T	F	F	F	F	T	T
F	T	T	T	F	T	F
F	T	F	T	F	T	F
F	F	T	T	F	T	T
F	F	F	T	F	T	T

Critical rows

Invalid argument : conclusion on 5th row is false!

Fallacy

A fallacy is an error in reasoning that results in an invalid argument.

Fallacy 1: converse error.

Example

- If it is Christmas, then it is a holiday.
- It is a holiday. Therefore, it is Christmas!

$$\begin{array}{l} p \rightarrow q; \\ \quad q; \\ \therefore\ p \end{array}$$

Fallacy 2: inverse error.

Example

- If it is raining, then I will stay at home.
- It is not raining. Therefore I would not stay at home!

$$\begin{array}{l} p \rightarrow q\ ; \\ \neg p; \\ \therefore \neg q \end{array}$$

Inference Rules

A rule of inference is a logical construct which takes premises, analyzes their syntax and returns a conclusion.

We already saw

$$\begin{array}{r} p \rightarrow q; \\ p; \\ \therefore q \end{array}$$

Modus ponens
(method of affirming)

$$\begin{array}{r} p \rightarrow q; \\ \neg q; \\ \therefore \neg p \end{array}$$

Modus tollens
(method of denying)

More Inference Rules

Conjunctive Simplification (particularizing)

$p \wedge q;$
$\therefore p$

Conjunctive Addition (specializing)

$p;$
$q;$
$\therefore p \wedge q$

Disjunctive addition (generalization)

$p;$
$\therefore p \vee q$

Disjunctive Syllogism (case elimination)

$p \vee q;$
$\neg p;$
$\therefore q$

Rule of contradiction

$\neg p \rightarrow C;$
$\therefore p$

Alternative rule of contradiction

$\neg p \rightarrow F;$
$\therefore p$

Inference Rule: Dilemma

Dilemma (case by case discussions)

$p \vee q;$
$p \rightarrow r;$
$q \rightarrow r;$
$\therefore r$

Inference Rule: Hypothetical Syllogism

Hypothetical syllogism:

$$\boxed{\begin{array}{l} p \rightarrow q; \\ q \rightarrow r; \\ \therefore\ p \rightarrow r \end{array}}$$

- Example

  If I do not wake up, then I cannot go to work.

  If I cannot go to work, then I will not get paid.

  Therefore, if I do not wake up, then I will not get paid.

Proof of Hypothetical Syllogism

$p \to q;\ q \to r;\ \therefore\ p \to r$

$(p \to q) \wedge (q \to r)$ (hypotheses; assumed true)

$\equiv (p \to q) \wedge (\neg q \vee r)$ (Conversion Theorem)

$\equiv [(p \to q) \wedge \neg q] \vee [(p \to q) \wedge r]$ (Distributive)

$\equiv [((p \to q) \wedge \neg q) \vee (p \to q)] \wedge [((p \to q) \wedge \neg q) \vee r]$ (Distributive)

$\equiv (p \to q) \wedge [((p \to q) \wedge \neg q) \vee r]$ (Recall absorption law: $a \vee (a \wedge b) \equiv a$, hence $[((p \to q) \wedge \neg q) \vee (p \to q)] \equiv p \to q$)

$\equiv (p \to q) \wedge [((\neg p \vee q) \wedge \neg q) \vee r]$ (Conversion)

$\equiv (p \to q) \wedge [((\neg p \wedge \neg q) \vee (q \wedge \neg q)) \vee r]$ (Distributive)

$\equiv (p \to q) \wedge [((\neg p \wedge \neg q) \vee F) \vee r]$ (Negation)

$\equiv (p \to q) \wedge [(\neg p \wedge \neg q) \vee r]$ (Unity)

$\equiv (p \to q) \wedge [(\neg p \vee r) \wedge (\neg q \vee r)]$ (Distributive)

$\equiv [(p \to q) \wedge (\neg q \vee r)] \wedge (\neg p \vee r)$ (Commutative; Associative)

$\therefore\ (\neg p \vee r) \equiv p \to r$ (Conjunctive simplification; conversion)

TABLE 1 Rules of Inference.

Rule of Inference	*Tautology*	*Name*
p $p \to q$ $\therefore q$	$(p \wedge (p \to q)) \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore \neg p$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive syllogism
p $\therefore p \vee q$	$p \to (p \vee q)$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \to p$	Simplification
p q $\therefore p \wedge q$	$((p) \wedge (q)) \to (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \to (q \vee r)$	Resolution

Inference rule: Resolution

$$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r).$$

Example: Show that the premises $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$.

Solution: We can rewrite the premises $(p \wedge q) \vee r$ as two clauses, $p \vee r$ and $q \vee r$. We can also replace $r \rightarrow s$ by the equivalent clause $\neg r \vee s$. Using the two clauses $p \vee r$ and $\neg r \vee s$, we can use resolution to conclude $p \vee s$.

- **Example:** Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."
- *Solution:* Let

  p ="It is sunny this afternoon,"

  q = "It is colder than yesterday,"

  r = "We will go swimming,"

  s = "We will take a canoe trip,"

  t = "We will be home by sunset."

  We need to construct an argument to show that the premises

  $$\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, \quad s \rightarrow t$$

  lead to the desired conclusion t.

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

- Example: Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."
- *Solution:* Let

  p ="You send me an e-mail message,"

  q = "I will finish writing the program,"

  r = "I will go to sleep early,"

  s = "I will wake up feeling refreshed."

  Then the premises are $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$.

  The desired conclusion is $\neg q \rightarrow s$.

Step	**Reason**
1. $p \to q$	Premise
2. $\neg q \to \neg p$	Contrapositive of (1)
3. $\neg p \to r$	Premise
4. $\neg q \to r$	Hypothetical syllogism using (2) and (3)
5. $r \to s$	Premise
6. $\neg q \to s$	Hypothetical syllogism using (4) and (5)

Exercise 16. Show that this argument is valid:

$$\boxed{\neg p \to F; \therefore p.}$$

Exercise 17. Show that this argument is valid, where C denotes a contradiction.

$$\boxed{\neg p \to C; \therefore p.}$$

Exercise 18. Determine whether the following argument is valid:

$$\begin{aligned}
&\neg p \to r \wedge \neg s \\
&t \to s \\
&u \to \neg p \\
&\neg w \\
&u \vee w \\
&\therefore t \to w.
\end{aligned}$$

Exercise 19. Determine whether the following argument is valid:

$$
\begin{aligned}
&p \\
&p \vee q \\
&q \to (r \to s) \\
&t \to r \\
&\therefore \neg s \to \neg t.
\end{aligned}
$$